ANALYTICAL REPORT

Diversification Across Six Major Crypto Assets

How much volatility an equal-weight holding actually removes, measured over 1,932 complete trading days

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Repository	crypto-sql-warehouse
Panel	1,932 complete trading days, 2021-06-19 to 2026-10-08, 6 assets
Source	`marts.daily_metrics` on PostgreSQL 17.10
Source of figures	`reports/metrics.json`, regenerated by `python scripts/run_diversification_analysis.py`

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1. Question and why it matters

Holding several assets instead of one reduces risk only to the extent that those assets move independently. This report measures how much of that reduction six major crypto assets actually deliver. The assets are ADA, BTC, DOGE, ETH, LINK, and SOL, all USD spot pairs from a single exchange. The portfolio holds them in equal weight, 16.67% each.

The reason to measure it rather than assume it is that the usual case for holding several coins rests on them being different assets with different drivers. If their daily returns move together, an equal-weight basket is close to a single position spread across 6 tickers, and the extra fees and operational surface of 6 holdings buy very little risk reduction.

> Every volatility, correlation, and diversification figure in this report is computed on one basis, the 1,932-day panel of days where all 6 assets have a clean daily return. Drawdown and total return are computed from the daily closes over the same window, as Table 2 states. Section 2 states how the panel is built and section 5 states what it cannot support.

On this panel the equal-weight portfolio has an annualized volatility of 0.7141, against 0.8171 for the average single asset. That is 12.6% of the volatility removed. If the same 6 assets had kept their variances with zero correlation between them, the portfolio would have come out at 0.3394, removing 58.5%. The portfolio captures 21.6% of the reduction that independence between the same 6 assets would have given.

Two caveats belong with that number rather than after it. The panel is 1,932 days, and at that length the ordering of individual pairs in the correlation matrix is supported only between pairs that sit far apart. And Figure 3 shows all 6 assets falling together, each at least 76.7% below its peak, so the 12.6% is measured partly through a joint selloff, which is when correlations are widely reported to run highest. Section 5 gives both in full.

2. Data and method

2.1 Source and grain

Every number comes from `marts.daily_metrics` in the warehouse this repository builds. That mart holds one row per asset per UTC trading day, rolled up in SQL from 320,142 hourly candles (the open, high, low, close, and volume for each hour) in `core.ohlcv` into 13,350 daily bars in `marts.daily_ohlcv`. Only the 1-hour rows of `core.ohlcv` feed that rollup. Daily bars are derived from hourly ones rather than fetched separately, so any disagreement between the two granularities can only be a rollup bug, which is findable, and never a source discrepancy, which is not.

The day boundary is pinned to UTC in the rollup expression rather than left to the session timezone. A plain cast would rebucket the whole mart for a client connecting from another zone, which is a bug this warehouse has already had once and now has a regression test for. The verify suite re-runs the mart's own defining query under America/Chicago and again under UTC and fails if any row differs, and the database default of UTC stays as a second layer.

2.2 The completeness rule, and why partial days are excluded

The loaded window starts and ends partway through a day, and the exchange has real outages, so some asset-days are built from fewer than 24 hourly candles. 39 of the 13,350 daily rows are partial in that way.

A partial day is a different measurement, not a smaller one. Its close is whatever the last observed hour printed rather than the 23:00 candle, and its volume is the sum of however many hours happened to land. Whether it is an up day or a down day depends on when the loader ran rather than on the market. Any variance or correlation computed over such a bar inherits that distortion. The mart therefore keeps the observed open, high, low, close, and volume on those days and sets every derived column, including `log_return`, to NULL. On any complete day with no complete day directly before it, the mart sets the return to NULL as well, because a return computed across a gap covers more than one day while still labeled as one day.

2.3 Building one panel

A covariance matrix has to describe one portfolio over one set of dates, so the panel keeps only dates where all 6 assets have a non-null log return. Of 2474 observed dates, 542 are excluded: 11 have at least one partial bar (2020-01-01, 2020-01-30, 2020-09-04, 2020-10-20, 2021-03-18, 2021-06-03, 2021-06-17, 2023-03-04, 2025-10-25, 2026-05-08, and 2026-10-09), 527 are dates on which not every asset has a daily bar, and 4 are complete days with no complete day directly before them, so they have no one-day return (section 2.2). Each exclusion removes the date for all 6 assets at once, so the panel is not lopsided toward any name. What remains is 1,932 days from 2021-06-19 to 2026-10-08, 11,592 asset-days with no holes.

The alternative, computing each correlation over whatever dates that particular pair happened to share, lets different entries of the matrix describe different periods. A matrix assembled that way is not guaranteed to be positive semidefinite, and the portfolio variance it produces can come out negative. Dropping 542 dates for everyone is the cheaper problem.

2.4 Returns, annualization, and covariance

Returns are logarithmic, ln(close today / close yesterday), because they add across periods and are the standard input to a variance estimate. Volatility is the sample standard deviation of those daily returns with the n minus 1 divisor, multiplied by the square root of 365. Volatility is stated per year because that is how it is conventionally quoted, which lets these figures be compared with published ones.

The 365 is deliberate. Crypto trades every calendar day including weekends and holidays, so a year holds 365 return observations. The square root of 252 that equity work uses counts exchange sessions, and a market with no sessions off does not have 252 of them. Annualizing this data on 252 would report every volatility in this report about 17% lower than it is.

The covariance matrix is the sample covariance of the same daily log returns, also with the n minus 1 divisor, scaled by 365. Portfolio volatility is the square root of w'Cw, the weight vector times the covariance matrix times the weight vector, with w set to 0.1667 in every position. The zero-correlation counterfactual replaces C with its own diagonal. Every asset keeps its variance exactly, and every covariance is set to zero. Negative covariances put a portfolio below that counterfactual, which is what a hedge is for, and `tests/test_diversification_metrics.py` checks that the arithmetic behaves that way. Every correlation in Table 1 is positive, so for the case at hand the zero-correlation portfolio is the reference the real number is measured against.

2.5 One basis, stated once

The warehouse also publishes `marts.pairwise_correlations_90d`, which reports 0.649 to 0.883 across its 15 pairs. That mart runs on a trailing 90-day window and has 90 shared days per pair, so it is a different window

on the same data and its numbers are not interchangeable with the ones below. Nothing in this report uses them.

3. Results

3.1 How closely the assets move together

All 15 pairs sit between 0.634 and 0.845, averaging 0.723. The least correlated pair is DOGE and SOL at 0.634 and the most correlated is BTC and ETH at 0.845. There is no pair in this set that a portfolio could lean on for independence. Figure 1 shows the matrix as a heatmap, and Table 1 gives its values.

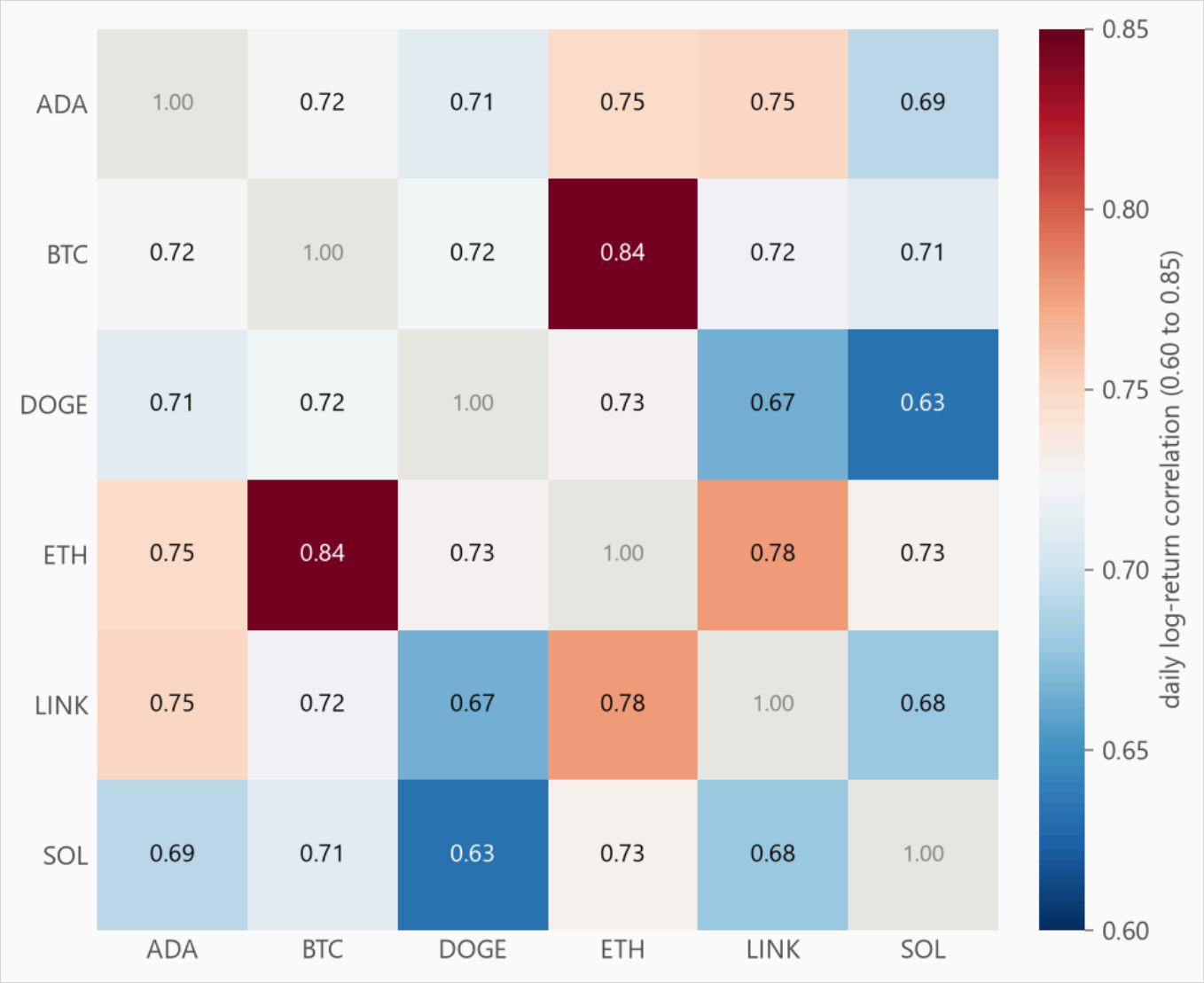


Figure 1. Pairwise correlation of daily log returns over the 1,932-day panel. The color axis runs from 0.60 to 0.85 rather than the full -1 to 1 range, because on a full scale every cell here would render in nearly the same tone. Read the colors as differences within a tight band, not as a general correlation scale. The diagonal is grayed out and left off the color scale, since a variable's correlation with itself is 1 by construction. The assets form one block, and the visible variation runs from 0.634 to 0.845, which is a band inside the high end rather than a range from low to high.

Table 1. The full correlation matrix behind Figure 1, on the 1,932-day complete-day panel.

	ADA	BTC	DOGE	ETH	LINK	SOL
ADA	1.00	0.724	0.711	0.747	0.750	0.691
BTC	0.724	1.00	0.722	0.845	0.721	0.711
DOGE	0.711	0.722	1.00	0.728	0.666	0.634
ETH	0.747	0.845	0.728	1.00	0.777	0.730
LINK	0.750	0.721	0.666	0.777	1.00	0.681
SOL	0.691	0.711	0.634	0.730	0.681	1.00

3.2 How volatile the assets are individually

Annualized volatility ranges from 0.529 for BTC to 0.977 for SOL, averaging 0.817. Figure 2 shows each asset against that average. The average is the reference the portfolio has to beat, because it is what an investor holding one of these names at random would have had.

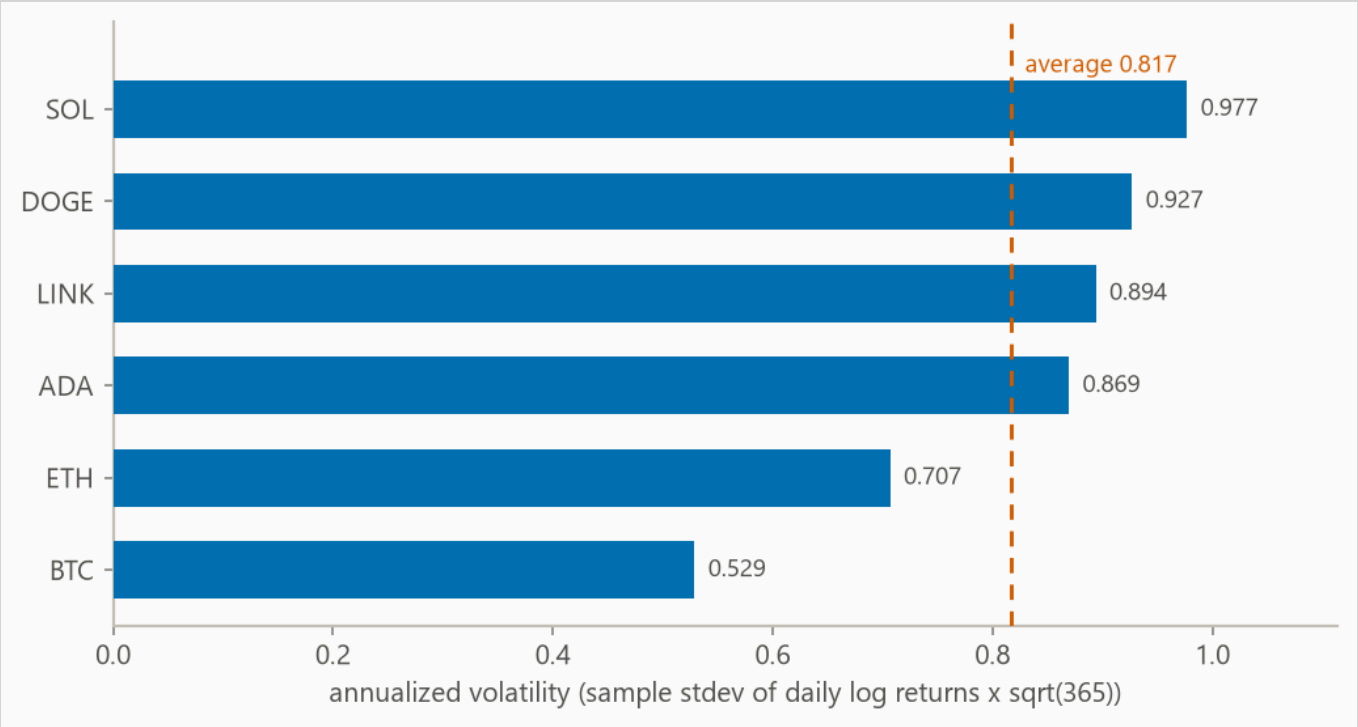


Figure 2. Annualized volatility per asset, sorted, with the 0.817 average marked. The assets are not interchangeable in risk terms. SOL has roughly 1.8 times the volatility of BTC, so equal weight is not the same thing as equal risk contribution.

Table 2. Per-asset figures. Volatility is computed from the panel's returns. Max drawdown and total log return are computed from the closes of every complete day from 2021-06-18 to 2026-10-08, so they include the moves on days the panel excludes. The first of those closes is the one the panel's first return is measured from. Drawdown is measured from the running peak within that window, not from an all-time high.

ASSET	ANNUALIZED VOLATILITY	MAX DRAWDOWN	TOTAL LOG RETURN
SOL	0.977	-96.3%	+1.090
DOGE	0.927	-85.2%	-1.251
LINK	0.894	-85.3%	-0.514
ADA	0.869	-95.2%	-1.809
ETH	0.707	-79.4%	+0.101
BTC	0.529	-76.7%	+0.824

3.3 What the co-movement looks like over time

The correlation matrix states the co-movement as 15 numbers. The drawdown paths in Figure 3 are consistent with it. The assets fall together and rebound together, and the differences between them are mostly depth rather than timing.

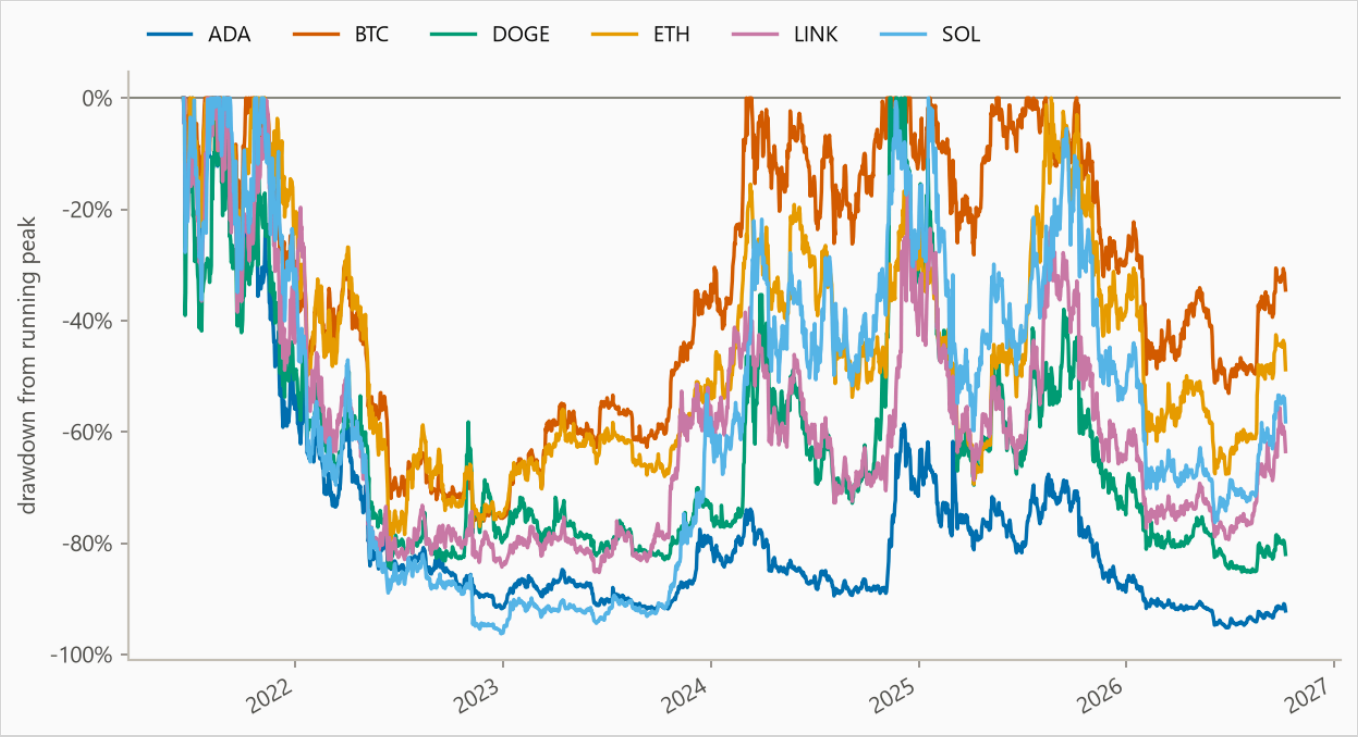


Figure 3. Each asset's drawdown from its own running peak, all on one axis. The assets fall together and rebound together. The deepest drawdown is -96.3%, for SOL.

3.4 The diversification result

Equal weights on the 1,932-day covariance matrix give an annualized portfolio volatility of 0.7141. The average single asset had 0.8171, so the basket removed 12.6% of the volatility. Holding the same 6 variances with every correlation set to zero would have given 0.3394, a reduction of 58.5%. The ratio of the two, 21.6%, is the share of the available diversification these assets actually delivered. Figure 4 puts the three volatilities side by side, and Table 3 lists them with the two reductions and their ratio.

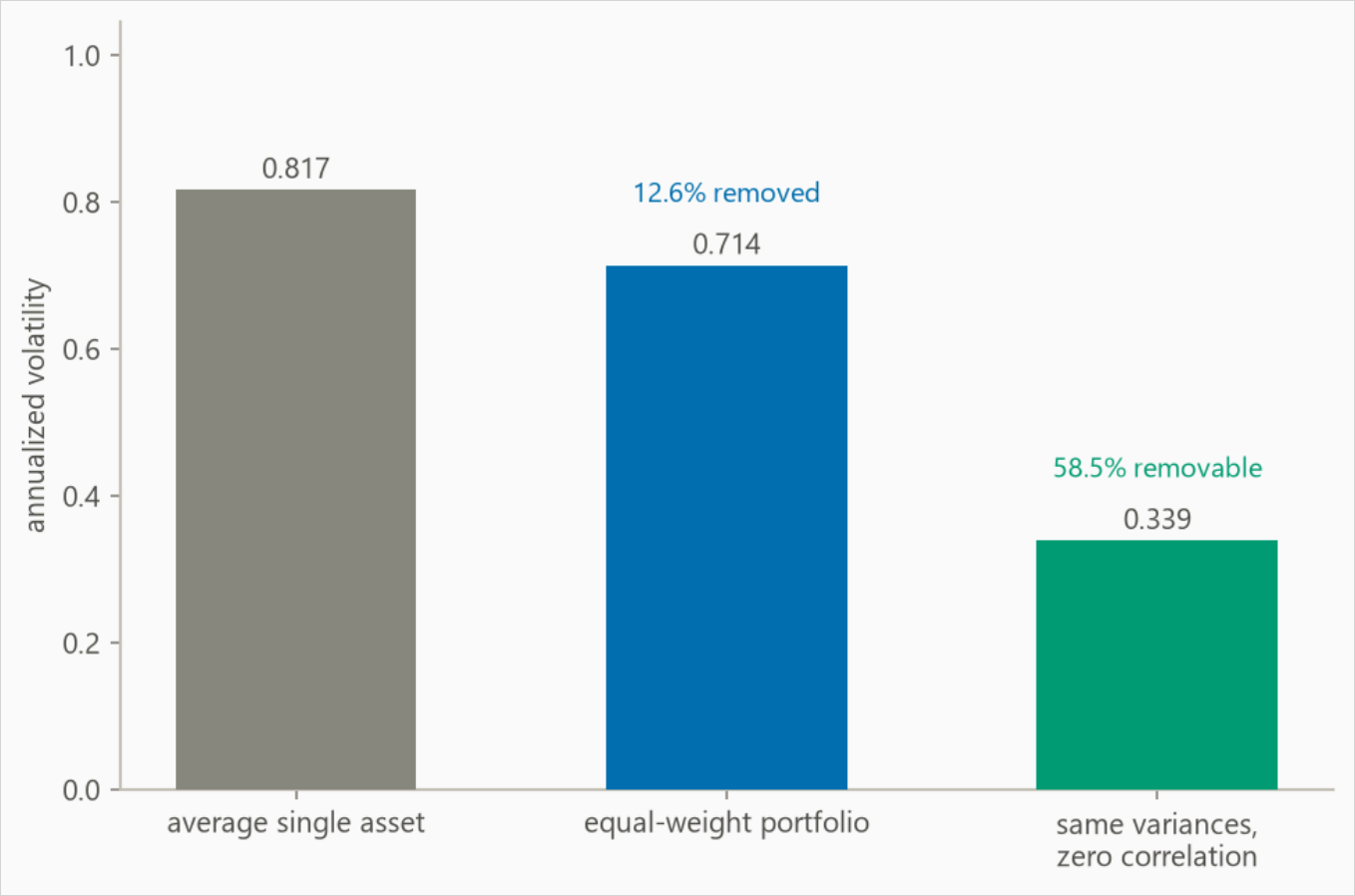


Figure 4. Three annualized volatilities: on the left the average of holding one of these assets, in the middle the equal-weight portfolio, on the right the same 6 variances with zero correlation. The left-to-middle gap is the benefit received and the middle-to-right gap is what the correlations consumed. Read the position of the middle bar. It sits close to the left one and far from the right one.

Table 3. The three volatilities, all annualized, with the two reductions and their ratio.

QUANTITY	VALUE
Average single-asset volatility	0.8171
Equal-weight portfolio volatility	**0.7141**
Same variances, zero correlation	0.3394
Risk removed by the portfolio	12.6%
Risk removable at zero correlation	58.5%
Share of available diversification captured	**21.6%**

4. What this means in practice

Spreading a holding across these 6 names removed 12.6% of its volatility. Counting names is therefore close to useless as a risk control here. Independence between the same 6 assets would have removed 58.5%, so the shortfall sits in the correlations and not in the number or the size of the positions. The actual correlations deliver 21.6% of that available reduction. An equal-weight basket of equally correlated assets converges toward the average volatility times the square root of the mean correlation. That limit is 0.695 here against the portfolio's 0.714, so another name from the same group would buy very little.

The sharper version of the same point is in Table 2. BTC alone had a volatility of 0.529, which is below the equal-weight basket's 0.714. Over this window, spreading into the other 5 names raised volatility rather than lowering it, because the diversification benefit was smaller than the extra variance those names brought with them. Equal weight assumes the assets are interchangeable, and on this evidence they are not.

Two consequences follow for anyone using this warehouse. First, position size and hedging do the work that name count does not. At a mean correlation of 0.723, total exposure is the variable that moves portfolio risk. Second, the quantity worth monitoring is the correlation level itself. `marts.pairwise_correlations_90d` recomputes it over the trailing 90 days at every refresh and keeps no history, so following it over time means saving a copy after each refresh. Only a fall in that correlation would change the conclusion here.

None of this is an argument about which assets to hold. It measures what the 6 names already loaded did over one window. Section 5 states what that window does not cover.

5. Limitations

The panel is 1,932 days. A single pair at the average correlation of 0.723 has a 95% Fisher interval of 0.701 to 0.743 on this many observations, a width of 0.043. Fisher's method keeps the interval inside -1 to 1 and accurate for correlations as high as these. The entire spread between the least and most correlated pair in Figure 1 is 0.211, so the uncertainty on any one correlation is about 20% as wide as the range the heatmap is showing. Two pairs' intervals stop overlapping only when their correlations differ by more than about 0.043, so this check supports the figure's ordering only between pairs that far apart. Beyond that, what survives is that the whole block sits high.

The data covers one exchange and one quote currency. All candles come from Coinbase and every pair is quoted in USD. There is no second venue to check the prices against, so a venue-specific outage or a stale feed enters the covariance matrix as if it were market structure. Common exposure to the dollar sits inside every one of these correlations, and with no second quote currency loaded there is no way to separate it out.

Correlations are not stable, and they are widely reported to rise in selloffs. This is the limitation that matters most for how the diversification result should be read, because a selloff is when diversification is being asked to do something. This report does not test that pattern. The panel holds more than one joint decline, visible in Figure 3, and in the deepest every asset fell at least 76.7% below its peak. No part of this analysis separates selloff days from the rest, so it cannot say how correlation inside a selloff differs from correlation outside one. The 12.6% is therefore a reading taken partly through a stressed market. If correlations do rise in selloffs, the days outside the declines lowered the measured correlation below its selloff level, so the 12.6% is more likely to be optimistic than conservative as a forecast for the next selloff.

Drawdown is measured only over the days this report covers. The peak in Table 2 and Figure 3 is the highest close since 2021-06-18, not an all-time high. Backfilling earlier history could change any drawdown figure in this report. These are not the numbers a trading desk means by maximum drawdown.

Equal weight is not an optimized portfolio. The weights are 0.1667 each because that is the simplest allocation to state, not because anything here selected them. A minimum-variance allocation cannot have a higher portfolio volatility than equal weight on this panel, because it is chosen to minimize exactly that. Given the 1.8 to 1 spread in single-asset volatility, a risk-parity allocation would probably have a lower one too. The w'Cw calculation also assumes the weights stay at 0.1667, which in practice needs rebalancing, and no transaction costs, funding, borrow, or slippage are modeled anywhere in this report.

Six assets is not the market. These are large-cap USD spot pairs, which is one corner of one asset class. I picked them for coverage and liquidity when I built the warehouse. They are not a sample of anything. A result about ADA, BTC, DOGE, ETH, LINK, and SOL is a result about those 6 names on this venue over these 1,932 days, and reading it as a claim about crypto in general would be reading it too widely.

6. How to reproduce

Reproduction needs Python 3.11 or 3.12 and a PostgreSQL server you can connect to. This run used PostgreSQL 17.10. Nothing in the SQL needs a feature newer than PostgreSQL 10. From the repository root (on Mac or Linux, use `cp` in place of `copy`):

```
pip install -r requirements.txt
copy .env.example .env
python scripts/run_setup.py
python scripts/run_ingest.py
python scripts/run_diversification_analysis.py
```

Between the copy and `run_setup.py`, edit `DATABASE_URL` in `.env` to point at your server. `run_setup.py` creates the database and applies the schema and the marts. `run_ingest.py` backfills candles at every active timeframe, which on a fresh setup is 1 hour alone, and refreshes the marts. The first run pulls the full backfill and every later run resumes from the watermark, the last candle already loaded.
`run_diversification_analysis.py` builds the panel, writes `reports/metrics.json` and the four figures under `reports/figures/`, prints the main results to the log, and then renders this document. When Chrome is installed at its default Windows location, the render also prints the document to PDF.
`run_build_report.py` repeats only the render, for a wording change that needs no new numbers. Every measured quantity in the prose above is read from `reports/metrics.json` at render time, so no number here can drift from the analysis that produced it. The fixed conventions the method rests on, the 365-day year and the 24-hour day, are written into the template and stated in section 2.

The pipeline underneath is checked three ways. `python scripts/run_verify.py` runs 7 checks against a loaded warehouse, each fatal on failure, covering candles missing from core that the loader never requested, rollup coverage, raw-to-core agreement, upsert idempotency, run status, session-timezone invariance of the daily rollup, and agreement of the two stored daily marts with their definitions. `pytest` runs 172 test functions, none of which need a database or a network. They cover the ingestion windowing and watermark, the API client, the record of each request to Coinbase, the re-request of unrecorded gaps, asset registration, the marts rebuild, the migrations, the CSV export, and the diversification math in this report, and one checks that every function is documented. The 21 annotated queries in `sql/04_queries.sql` each run standalone against a loaded database.

Rerunning the ingestion loads whatever the exchange has published since, which moves the panel's end date and every number here. This run's history covers 2,474 dates. A fresh first load reaches back 365 days, so it gives a shorter panel and different numbers. Passing `run_ingest.py` a `--backfill-days` value that reaches back past 2021-06-18 loads enough history for this panel. The figures in this document are the ones produced by the run recorded in `reports/metrics.json`, generated 2026-10-09T11:40:08Z against PostgreSQL 17.10.

Generated from `reports/metrics.json` by `src/crypto_warehouse/report_generator.py`. Panel 2021-06-19 to 2026-10-08, 1,932 complete days, 6 assets, 11,592 asset-days.